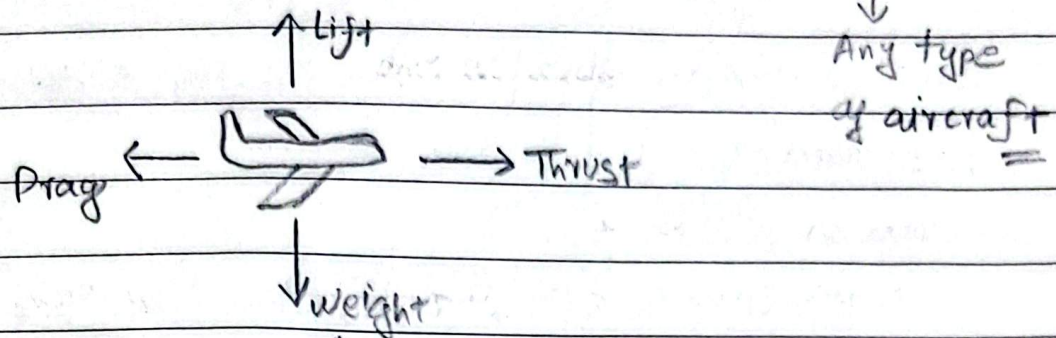


Aerodynamics of Flight - The forces of flight

There are primarily four forces that act upon an aircraft.



* All four forces must be balanced

To maintain a straight & level flight*

Gravitational pull.

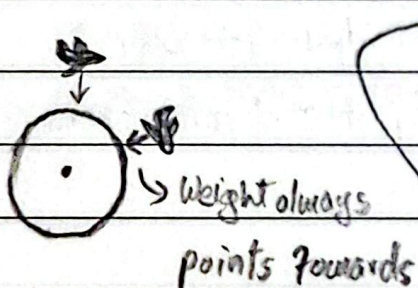
Feibl force

Vector Quantity

$W = g \times m$ (Simplified formula to calculate weight).

First Challenge of aviation:-

How do we overcome the weight of the aircraft with some other force to achieve ~~stained~~ sustained flight?



Second challenge of aviation:-

How do we control the aircraft during flight to make these adjustments?

the centre of gravity.

Changes throughout the course of flight must be countered by making adjustments.

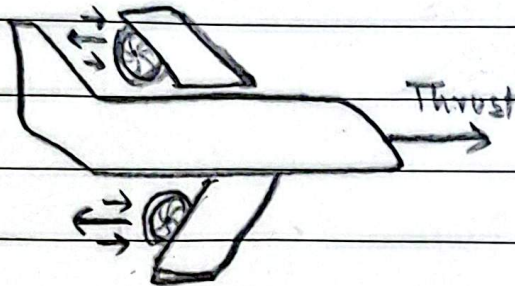
Note these two challenges. To solve them, we need to look at the following information.

Newton's 3rd law $\rightarrow$ Every action must have equal & opposite reaction.

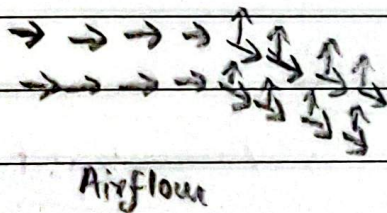
Similarly, $\rightarrow$ rocket

A ~~jet~~ engine burns a propellant and shoots it back at extreme speeds. This analogy can be

extended to jet engines, where air is pushed back, generating thrust.



Newton's 1st law: A body at rest will remain at rest & a body in motion will ~~remain~~ continue in a straight-line motion unless subjected to an external applied force.



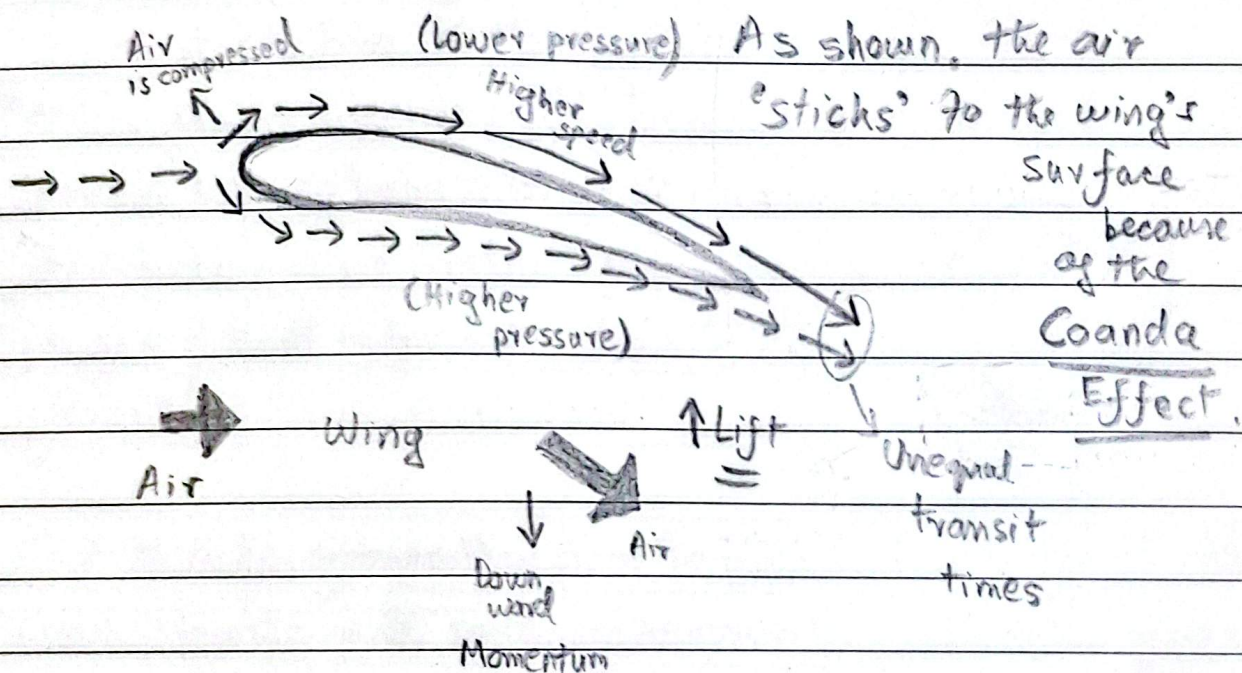
$\rightarrow$ According to N1, if airflow deflected, an external force was applied.

* So we see that directing airflow downwards helps us produce lift.*

$\rightarrow$ According to N3, if airflow is downwards, there must be an upwards force. (Lift)

Wings & Generation of Lift

Bernoulli Effect - Higher pressure below the wings pushes the wing upwards, while lower pressure above the wing pulls it upwards.



Conclusion: Lift is generated as a result of N1, N3 and the pressure differing theory. (Bernoulli Effect).

The Lift Formula

$$\text{Lift} = (C_L) (S) \left(\frac{\rho}{2}\right) (v)^2$$

coefficient of lift

Surface area of wing

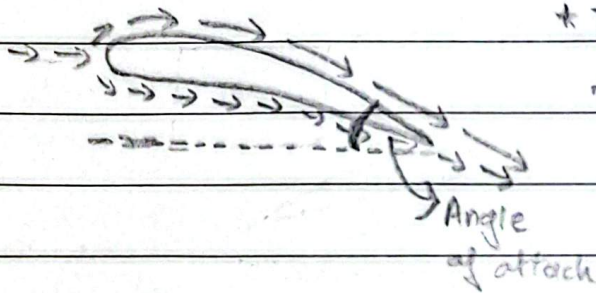
velocity of air

How to increase lift generation

Method 1: Higher speed

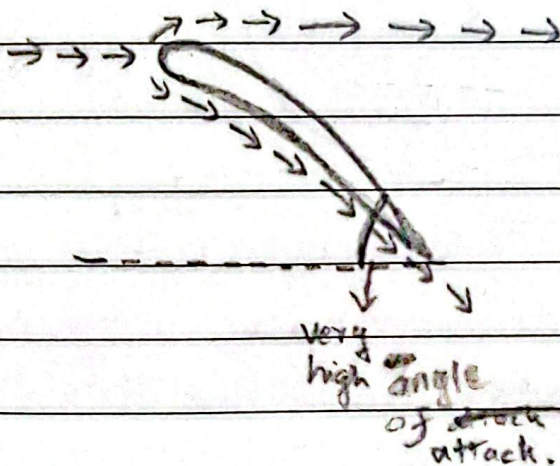
- * Magnitude of deflected air increases.
- * Increasing the magnitude of the lift vector consequently.

Method 2: Increasing angle of attack



- * Tilt the wing more upwards relative to the direction of the air flow.

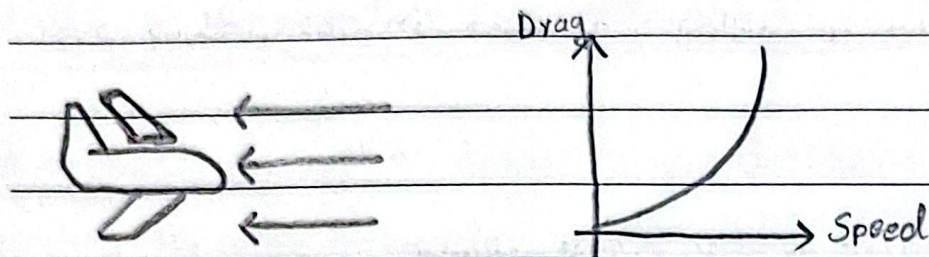
- * Extreme increase in the angle of attack has a huge problem*



Stalling

When the air no longer 'sticks' to the wing due to an extreme angle of attack. Causing loss of lift.

Aerodynamics of Flight 2 - The types of Drag



This is parasite drag. It's the resistance offered by the air to anything moving through it. It can be also called Aerodynamic Friction.

This drag force has an exponential relationship with speed & increases with the square of speed.

Subtypes of parasite drag :-

Skin Friction Drag →

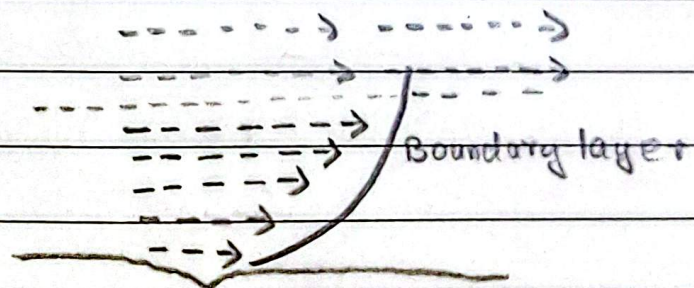
* Surface may appear smooth, but

it inevitably has dents which cause pockets of air to form*

Aerodynamic resistance due to contact of air with respect to surface of the aircraft.

Form Drag (Next Page)

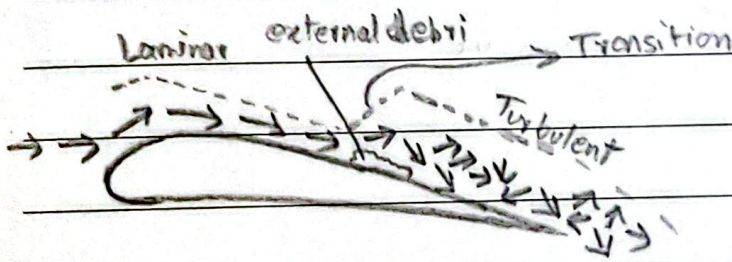
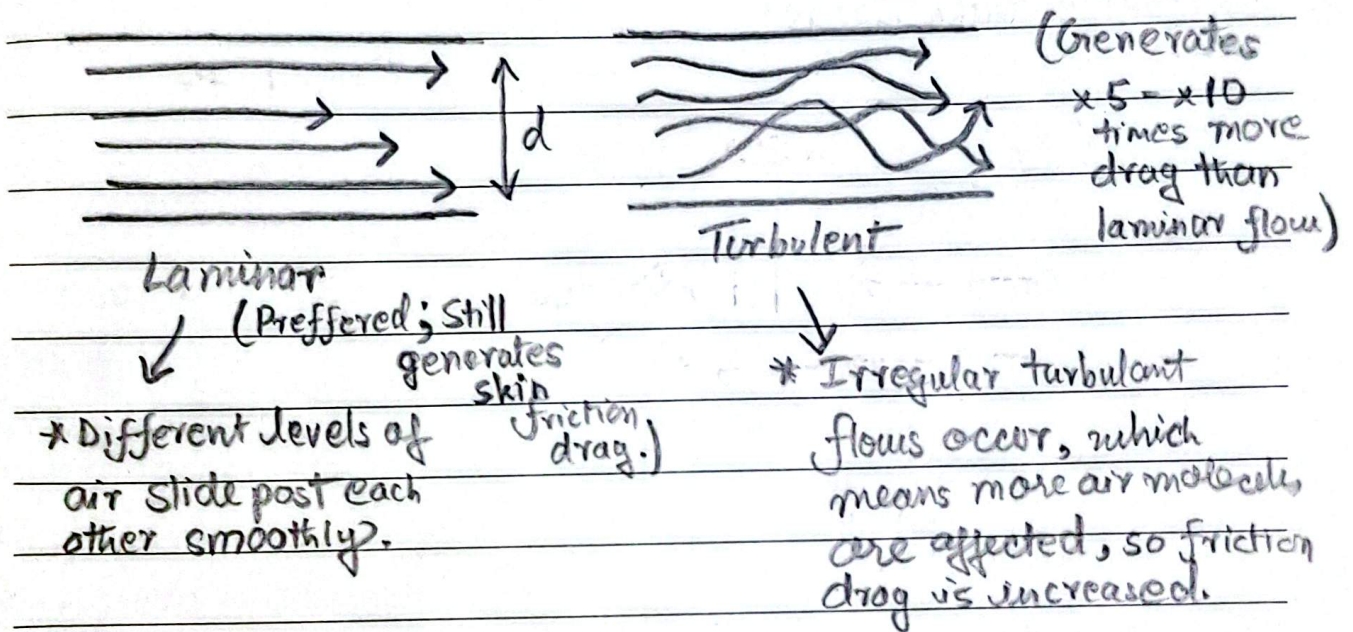
Interference Drag (Next Page)



Air is viscous. The stationary air molecules created by skin friction drag slows the layer above it, and continued but a bit less each time. These layers of slowed down air is called the boundary layer.

The viscous resistance retards forward motion, thus producing drag.

There are two types of boundary layers:-



Turbulent flow can be caused by bumps & imperfections on a plane's surface.

So, to reduce or delay this

transitions, the following steps can be taken:-

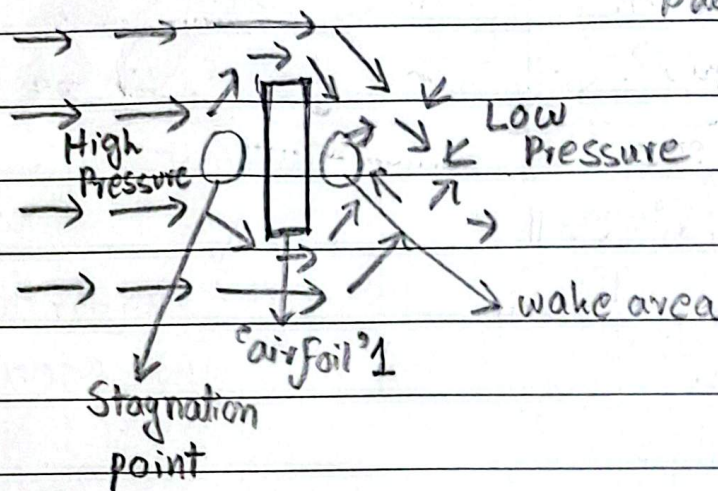
- * Waxing
- * Cleaning
- * Special coating agents

Form drag & interference drag is discussed afterwards.

Form Drag

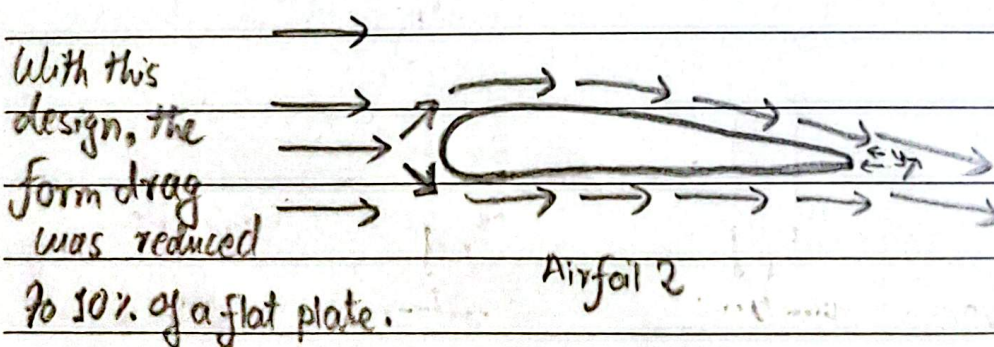
Form drag results from the turbulent wake caused by the separation of airflow from the surface of a structure.

Unstreamlined airfoils form stagnation points & low pressure wake areas.



* Due to the Bernoulli Effect, the zone of high pressure pushes back against the area of low pressure*

To reduce form drag, airfoils must be streamlined.



Another method to reduce drag is to reduce the surface area of the airfoil that comes in contact with air.

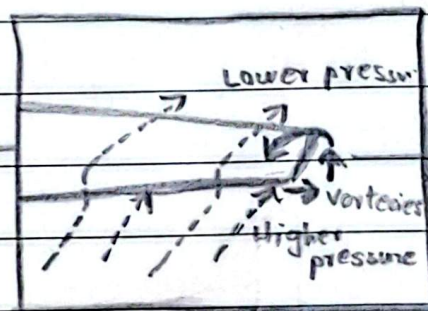
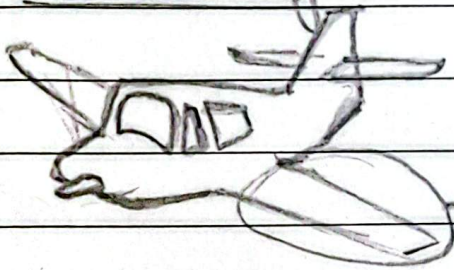
Interference Drag

Interference drag occurs when varied currents of air meet & interact. This drag can occur between several components of an aircraft, at places where they are connected, where different air currents meet.

Strategic placements of fairings & fillets, which use curved surfaces to allow for airflow streamlines to meet gradually, ease the transition between components & reduce interference drag.

So, parasite drag has three subtypes, Skin friction, form & interference.

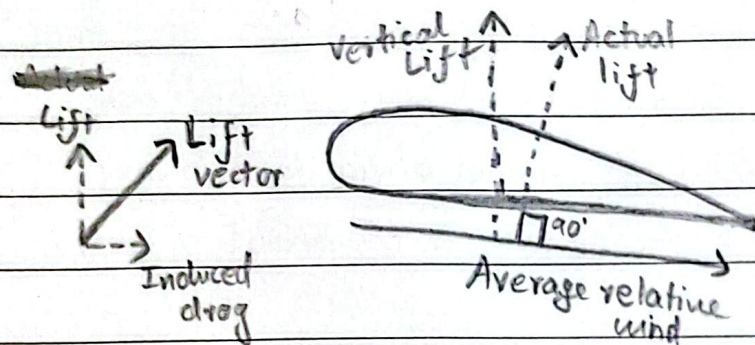
Induced Drag



Induced drag is an inevitable by-product of generating lift, generated when a wing is driven through air to produce differing pressures. (Bernoulli Effect)

At the wingtip, air from higher pressure travels up to lower pressure. The 'tip' effect is in more detail next page.

The vortex's starting upwards motion is from the tip while the ending downwards motion is at the extremity of the wingspan. This causes the net direction of airflow past the wing to be facing downwards.

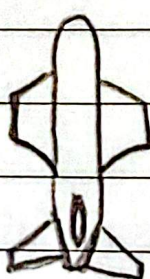


As shown, the lift vector being tilted slightly backwards produces drag.

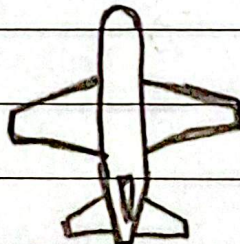
There are two main ways to reduce induced drag.

→ Increasing the aspect ratio of the wing

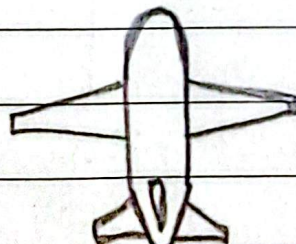
→ Reduce the air quantity upwards at the tip through wing twisting.



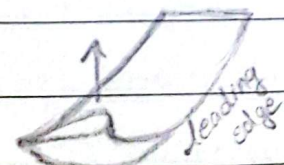
small aspect ratio



Medium aspect ratio

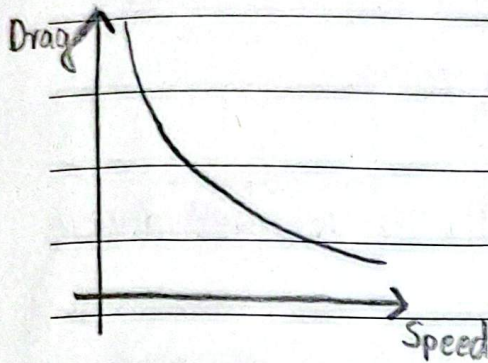


High aspect ratio



Wing twisting

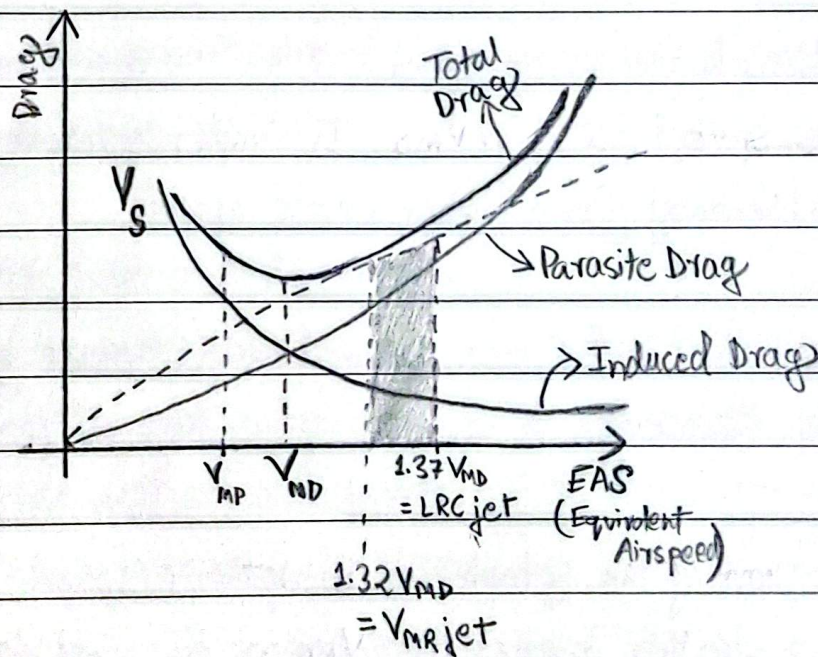
Induced drag is inversely proportional to the square of speed.



Notice how this is the complete opposite for parasite drag.

Total Drag

Total Drag is the sum of both parasite & induced drag.



* Points & respective meanings of the points on the graph is mapped out on the next page.*

① V_{MD} (Min. Drag Speed) $\rightarrow$ * Occurs at the lowest point of the total drag curve.

* Induced Drag = Parasite Drag

* V_{MD} provides the maximum Lift-to-Drag ratio.

② V_{MP} (Min. Power Speed) $\rightarrow$ * At this speed, the required power to maintain level flight is at its lowest.

③ V_{MR} jet (Max. Range Speed) $\rightarrow$ * This is a standard jet cruise speed and is at $\times 1.32$ the V_{MD} .

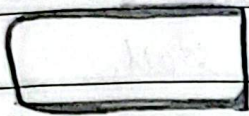
④ LRC jet (Long Range Cruise Speed) $\rightarrow$ * This is a standard jet cruise speed at $\times 1.37 V_{MD}$. It trades a small amount of fuel efficiency for a higher cruise speed.

⑤ V_S (Stall Speed) $\rightarrow$ * The min. airspeed which must be maintained in order for the wings to continue generating lift.

The shaded region of the graph is an optimal speed for flight. It is slightly higher than V_{MD} as if speed decreases at V_{MD} , drag increases, which further increases drag, causing V_S to approach rapidly and risking a stall.
(as aircraft slows down)

Aerodynamics of Flight 3 - Wing Planforms

Wing Planforms are the shape & layout of the fuselage & wing of a fixed-wing aircraft.



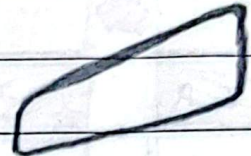
Rectangular
Wings



Elliptical
Wings



Tapered
Wings



Sweptback
Wings

There is a nearly endless variety of wing planforms designs in the history of aviation used in both experimental & production aircrafts. These are just a few.

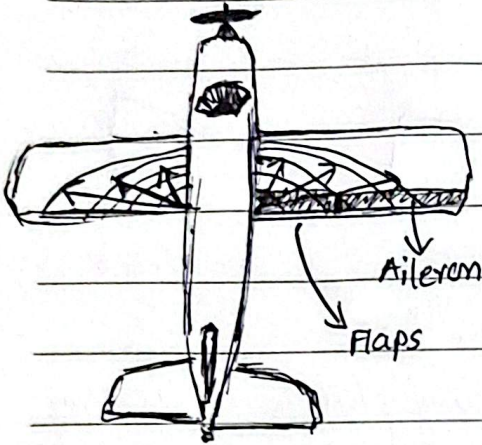
→ These are non-tapered, straight wings, extends out of the fuselage at 90° .

- | | |
|------------------------------------|-----------------------------------|
| - Easier & cheaper to manufacture | - Not aerodynamically efficient. |
| - Easy maintenance | - Produces a lot of induced drag. |
| - Great Strength | - Low speed wing |
| - Higher lift at low speeds | |
| - Strong maneuvering ability | |
| - Desirable stall characteristics. | |

(Explained in
next page)

These wings are best used in smaller aircrafts, such as cessna's, piper's or others such as agricultural-use aircrafts.

Desirable Stall Characteristics



Rectangular wings stall from the root, which are less severe and can warn the pilot of an impending stall.

Stalling from the root causes the ailerons & flaps to be affected last, making the aircraft controllable.

Elliptical wings

This is a wing whose leading and trailing edges resemble those of an ~~egg~~ ellipse.

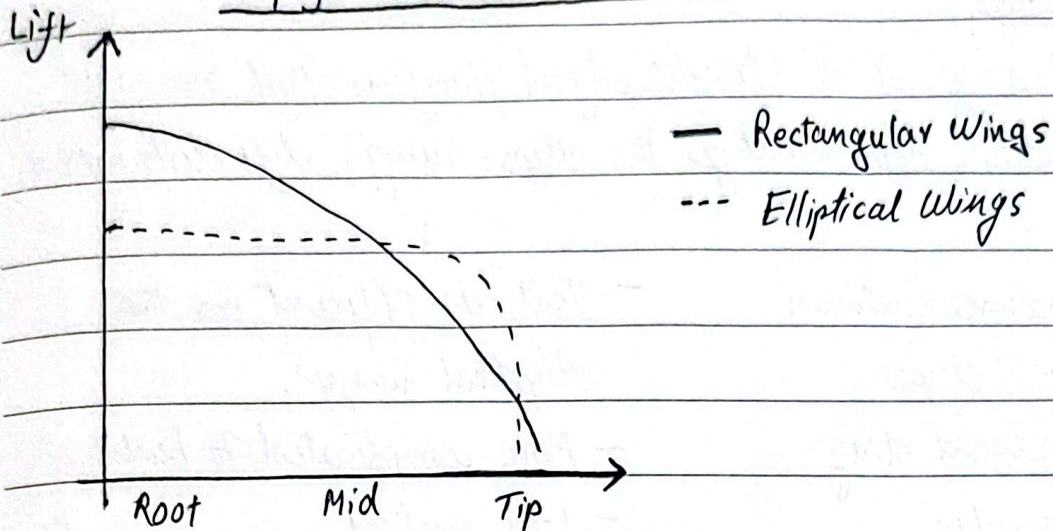
→ Very light
→ Reduced / lowest possible induced drag (small tips).

→ Difficult to manufacture due to compound curves.
→ Less rigid.

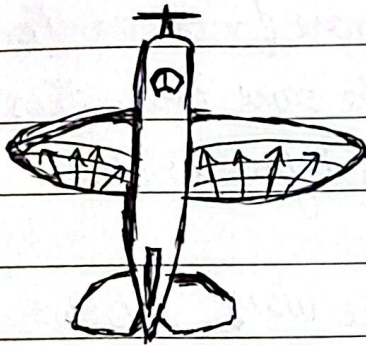
→ Undesirable stall characteristics.



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Simplified Lift Distribution curve

We can see that for the rectangular, the lift steadily decreases as we reach the tip. So the lift mostly comes from the root of the wing. However, for the elliptical wing, lift is fairly uniform until right before the tip.



So, uniform lift distribution leads to the entire ~~aircraft~~ aircraft stalling at the same time, with little to no warning.

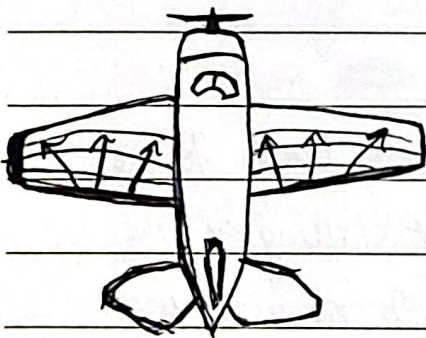
^{comparison}
This should clarify the differences between desirable & undesirable stall characteristics.

Elliptical wings are not commonly used, but were used majorly by WW2 ~~or~~ fighter jets.

Tapered Wings

The tapered wing is a straight edged planform that narrows towards the tip, approximating the ellipse wing's lift distribution.

- Good balance between rectangular & ellipse.
- Lower induced drag than rectangular.
- Better manufacturability compared to ellipse wing.
- Isn't as efficient as the elliptical wing.
- More complicated to build.
- Less control
- Less desirable stall characteristics.



As the wing becomes more tapered, the wing will stall more & more on its entire surface at the same time, thus having less warnings before a stall.

This wing was also greatly used during the WW2, notably the P-51 Mustang.

Swept-back wings

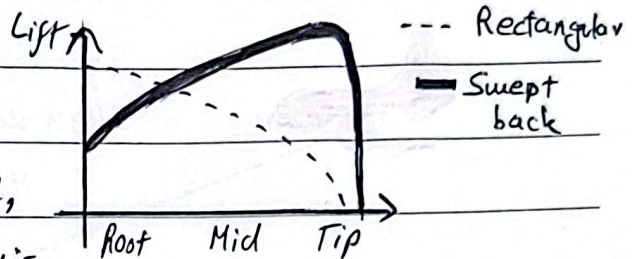
This is a wing whose leading edges are swept-back.

- Reduced drag in transonic flights.

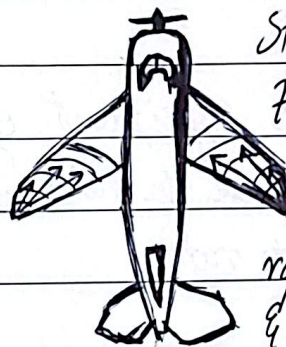
- Undesirable stall characteristics.

Mach 1: Speed of sound.

As the air flows over a wing surface, the air accelerates. On a airfoil we:-



Critical Mach No.:- Speed of aircraft at which air flowing over the wings reaches Mach 1.



Stalling starts from the tip, which cuts off aileron control.

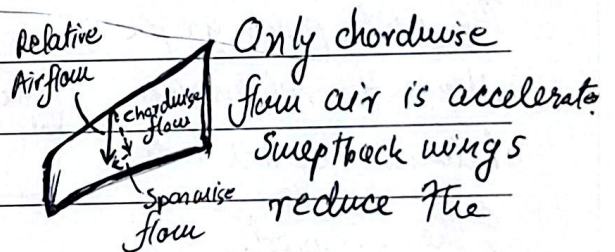
This can cause a rapid loss of control & can lead to a crash.

These wings are used in most high-speed commercial jets today.

When air slows down, the point at which air goes below mach 1, forms a shockwave, creating wave drag.



Air past the shockwave becomes turbulent, as it is tripped. The air separates from the wings, causing massive loss of lift.



Only chordwise flow air is accelerated. Sweptback wings reduce the chordwise (as it's now a vector of the total airflow), thus increasing Mach & delaying wave drag.

Glide Ratio

Glide Ratio is the distance of forward travel divided by the altitude lost.

$$\text{glide ratio} = \frac{\Delta \text{distance}}{\Delta \text{altitude}} \quad \text{Alternatively,}$$

$$\text{glide ratio} = \frac{\text{Airspeed (ms}^{-1}\text{)}}{\text{Sink Rate (ms}^{-1}\text{)}}$$



When all four forces of flight are balanced, the glide ratio stays constant.

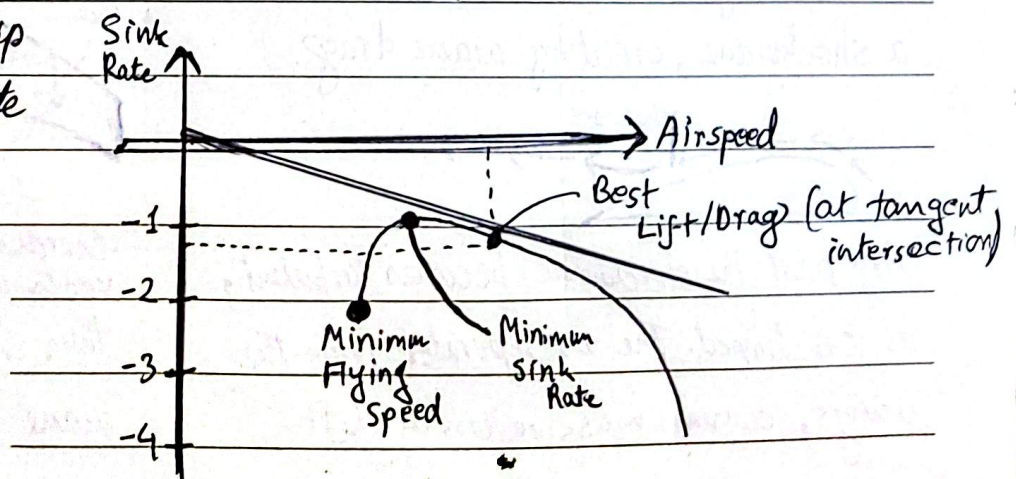
Wind velocity has significant effect on glide ratio. A tailwind will greatly increase the ratio whereas a headwind will greatly decrease the glide ratio.

Sink Rate

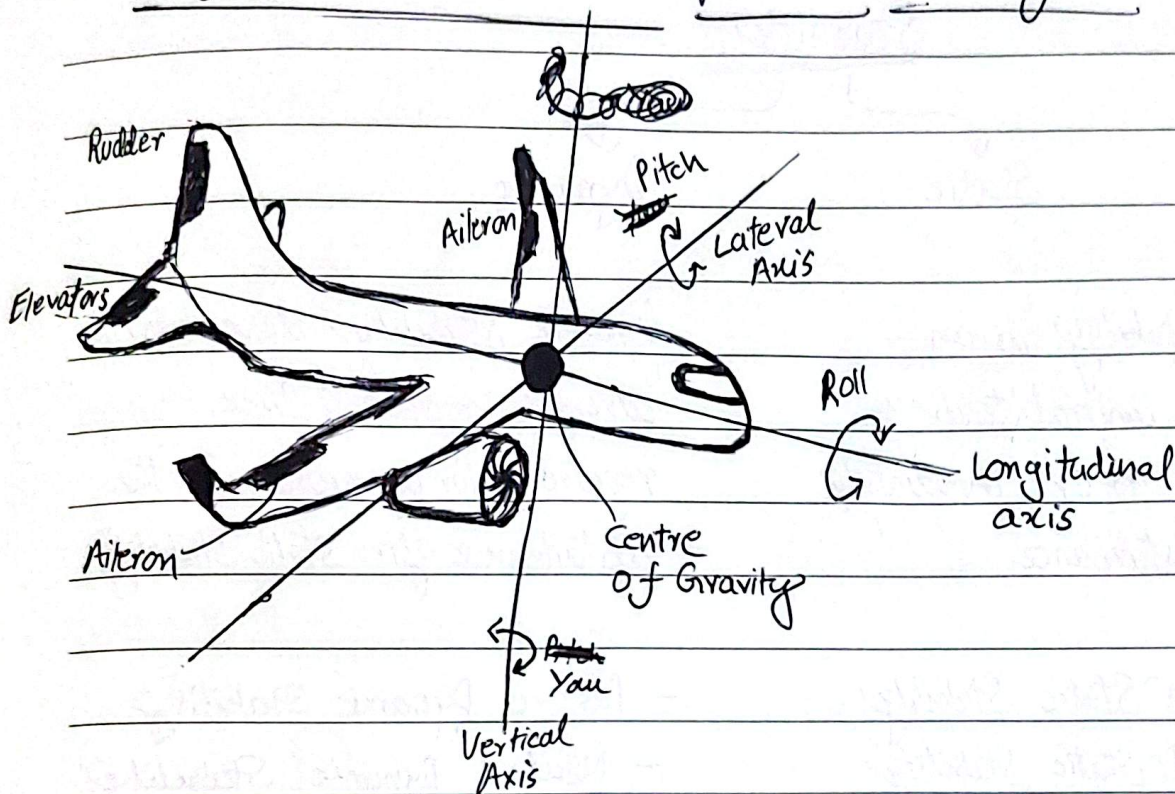
The sink rate is the change in altitude over time.

The glide polar graph shows the relationship between the sink rate & the airspeed.

$$\text{sink rate} = \frac{\Delta \text{Altitude}}{\Delta \text{Time}}$$



Aerodynamics of Flight 4 - Axes of Rotation & Stability



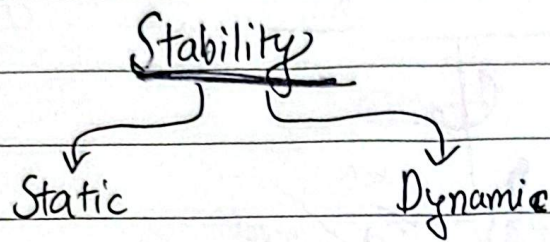
A fundamental principle of physics is that all rotation must happen about the point of the centre of gravity.

- * Ailerons control the roll (Rotation in the longitudinal axis)
- * Rudder controls ~~rotation~~ yaw (Rotation in the vertical axis)
- * Elevators controls ~~rotation~~ pitch (Rotation in the lateral axis)

Stability

Stability is an aircraft's ability to maintain a uniform flight condition & return to that condition after being disturbed.

More on stability in the next page.



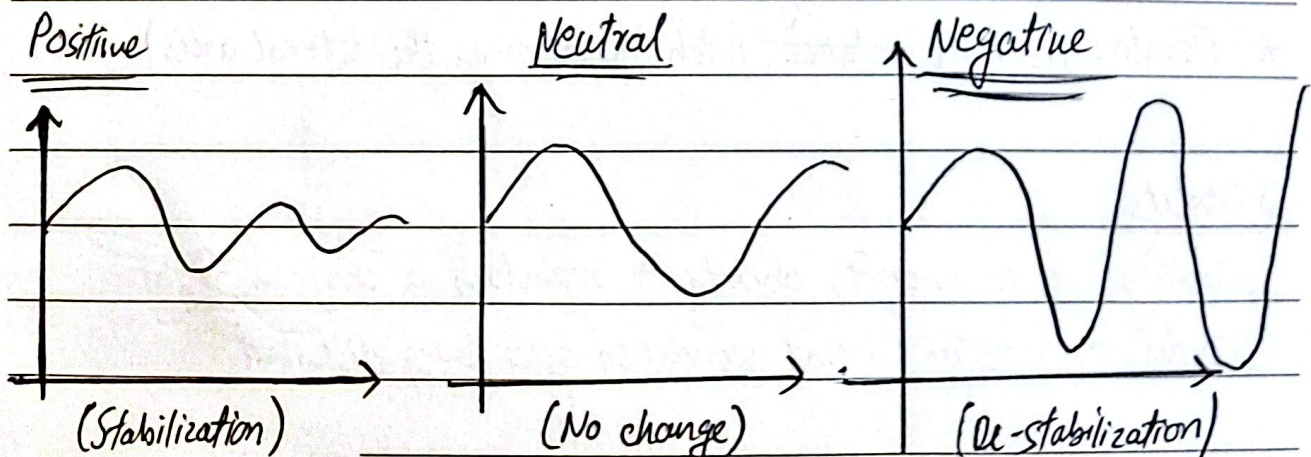
Static stability is an aircraft's initial tendency to return to eq^m immediately after disturbance.

Dynamic stability describes an aircraft's motion & time required for a response to the disturbance after static stability.

- Positive Static Stability
- Neutral Static Stability
- Negative Static Stability

- Positive Dynamic Stability
- Neutral Dynamic Stability
- Negative Dynamic Stability

To visualize, here is flight path representations after an aircraft gets hit with a disturbance.

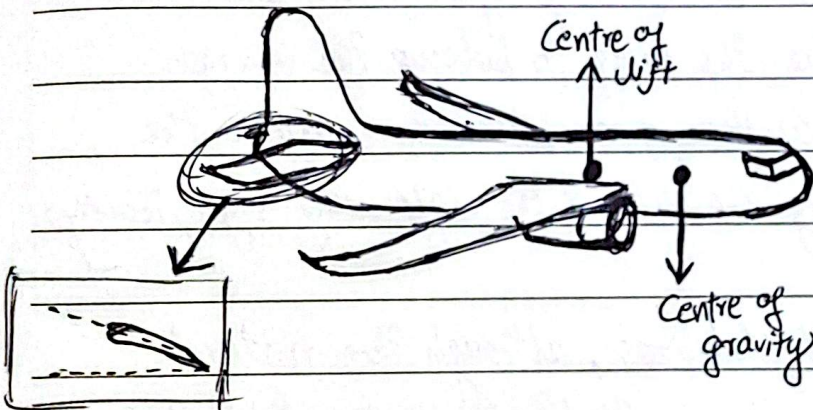


Both types of stability are particularly important for pitch control about the lateral axis.

The stability discussed about so far can be also called longitudinal stability. This is the stability about the longitudinal axis.



To ensure the aircraft has positive stability is to make the aircraft slightly nose-heavy.

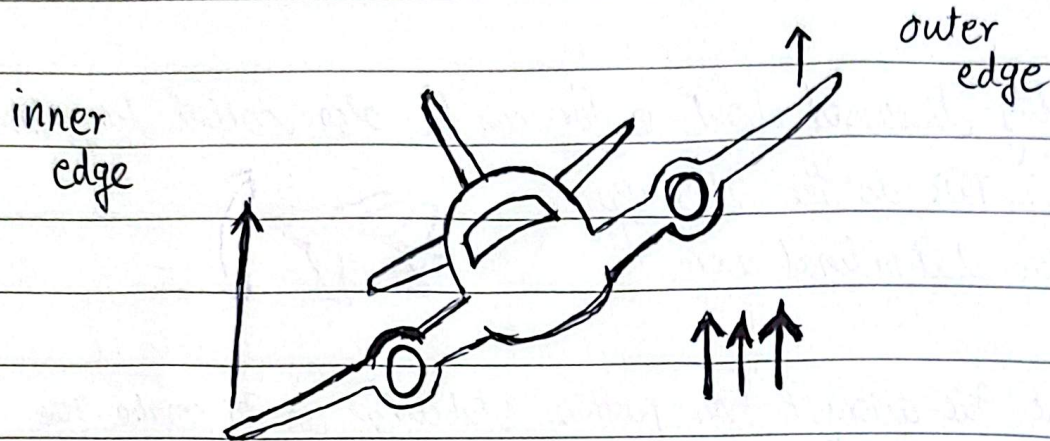


To offset this difference & nose-dive tendency, the angle of attack of the horizontal stabilizer has a negative AOA.

This way, when the airplane pitches, the amplitude of the oscillations should steadily decrease until the horizontal stabilizer perfectly offsets the tendency to dive.

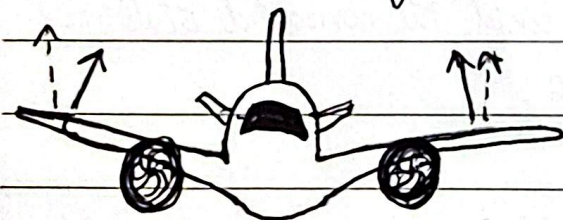
Next page looks into lateral stability, which is significant for roll control over the longitudinal axis.

Lateral Stability



Assume the aircraft enters a roll due to an updraft as shown. The wing on the outer edge is moving upwards, the relative air flow comes from above the wing to below the wing. The inner edge wing gets air flow from below the wing to the top of the wing, increasing the lift of the left wing significantly.

This means, the aircraft self-stabilizes, although this natural effect of the wings is not enough to ~~truly~~ truly stabilize the aircraft. This is why dihedral is built into aircraft wings.



--- Represent the effective lift of the wings, the ~~lift~~ vertical vector of total lift.

In the event of a roll, the horizontal component of the inner wings ~~decreases~~ decreases and the outer wings increases,

further exaggerating the previously shown 'natural effect'.

Shifting the ailerons also control the roll manually?



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CFD - Overview

- * Grid Generation
- * Fluid Dynamics
- * Boundary Conditions
- * Numerics

CFD is numerically solving an equation or sets of equations that govern fluid motion.

- * Visualization
- * Turbulence Modeling

CFD - Steps

- * Refine Problem
- * Pre-process $\rightarrow$ Setting up the problem
- * Solver $\rightarrow$ Solving the equations
- * Post-process $\rightarrow$ Analysis of the results.

Equations of Motion & Discretization

- * System of partial differential equations that govern fluid dynamics.
- * For given boundary condition & domain the system of PDE do not have analytical solutions.
- * These equations must be solved computationally.

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho u_j}{\partial x_j} = 0 \quad (\text{Conservation of mass})$$

Density ρ , time t , velocity u_j , spatial position x_j .
 $j=1$ (x-direction) / $j=2$ (y-direction) / $j=3$ (z-direction)

$\left[\frac{\partial \rho}{\partial t} = \text{This notation means partial derivative of } \rho \text{ w.r.t. time.} \right]$

$$\frac{\partial \rho u_i}{\partial t} + \frac{\partial \rho u_i u_j}{\partial x_j} = - \frac{\partial p}{\partial x_j} \delta_{ij} + \frac{\partial \tau_{ij}}{\partial x_j} \quad (\text{Conservation of momentum})$$

Static thermodynamic pressure p , Stress tensor τ_{ij} , Kronecker Delta Function δ_{ij} .

$$\frac{\partial \rho e_0}{\partial t} + \frac{\partial \rho u_j e_0}{\partial x_j} = - \frac{\partial u_j p}{\partial x_j} - \frac{\partial q_j}{\partial x_j} + \frac{\partial u_j \tau_{ij}}{\partial x_j} \quad (\text{Form of energy eqn})$$

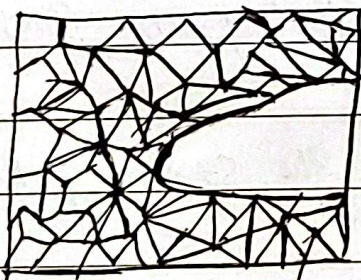
energy e_0 , Heat transfer term q_j .

(1st law of thermodynamics)

Pre-Processing

Computational Grid Generation

*



* Grid ~~number~~ points/nodes connect together to form a grid.

* As grid points are added to the solutions, the solutions should converge.

* The more ^{the} grid points, the better the results, but also require ~~by~~ huge computational power & the solver needs to run for longer.



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Solver - Convergence & Stability

- * Stability - The property of the solver where as it progresses towards finding the solutions of the discretised equations the error delays.
- * Consistency - A finite difference representation of a PDE is considered consistent if the difference between the PDE and its discretized form representation vanishes as the mesh is refined.
- * Convergence - The discretized form of the equations of motions should approach that of the governing equations as the mesh is refined and the solver progresses.

Convergence = Consistency + Stability

Post-Processing - Inspection of Solutions

Analysis, examination, plotting, visualisation, & extraction of CFD solution,

Graphical or other representations for understanding the physics & implication of the results.

Process that derives quantitative or qualitative flow physics from CFD solutions.

Numerical database containing the approx. sol. of discretized eqⁿs